



ACCREDITATION CRITERIA

1.0 PURPOSE OF THIS DOCUMENT

The purpose of this document is to describe the nine accreditation criteria that are the basis for evaluating a program by the Board of Accreditation for Engineering and Technical Education (BAETE) of the Institution of Engineers, Bangladesh (IEB).

2.0 SCOPE OF THIS DOCUMENT

The accreditation criteria presented here apply to the Higher Educational Institutions (HEI) offering undergraduate engineering programs in Bangladesh.

3.0 ROLES AND RESPONSIBILITIES

3.1 Engineering programs applying for BAETE Accreditation

- Prepare a Self-assessment report (SAR) based on the ten accreditation criteria. The Self-assessment report template for the programs must be prepared as per ACC-TMP-04-04 (V2.1 – effective till 30 June 2026; V3.0 – effective from 1 July 2025; programs may apply using template V2.1 or V3.0 between 1 July 2025 and 30 June 2026).

3.2 BAETE

- Conducts the program evaluation based on the accreditation criteria presented in this document.

4.0 DEFINITIONS AND ACRONYMS

See ACC-MAN-06 for *definitions and acronyms*.

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This section presents the nine accreditation criteria, which are the basis for evaluating a program for accreditation. The following subsections sequentially outline the required attributes of each criterion in terms of sub-criteria. This section does not provide any quantitative benchmark requirement to satisfy a criterion or sub-criterion adequately or appropriately. Adequacy or appropriateness is to be decided qualitatively, considering what is necessary for the students to attain the outcomes. It should be noted that no sub-criterion is assigned any weight. Each criterion must be holistically evaluated in terms of the qualitative benchmark requirements. The report template that the evaluation teams shall use is given in ACC-TMP-04-05.

5.1 Program Educational Objectives

Program Educational Objectives (PEOs) are broad statements describing the career and professional accomplishments the program prepares graduates to achieve. PEOs are assessable based on the attributes and accomplishments of graduates, preferably those who have worked for 3 to 5 years after graduation. Requirements under this criterion are described in terms of the following sub-criteria.

- i. Published PEOs must be clear, concise, assessable, and realistic within the context of the available resources.
- ii. PEOs must be consistent with the vision and mission of the institution or the department offering the program.
- iii. The program must demonstrate an established system for compiling the level of attainment of PEOs, including a mechanism for tracking and obtaining feedback from graduates and their employers.



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5.2 Program Outcomes and Assessment

Program Outcomes (POs) or graduate attributes are narrower statements describing what students are expected to know and be able to do by graduation. These statements relate to the knowledge, skills, and attitudes students acquire while progressing through the program. By graduation, the program must demonstrate that students have achieved an acceptable minimum level of specific knowledge, skills, and behavioral traits. The BAETE requires that students acquire the following POs or graduate attributes:

- PO1. Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.
- PO2. Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
- PO3. Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5).
- PO4. Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
- PO5. Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
- PO6. When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).
- PO7. Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).
- PO8. Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9).
- PO9. Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences.
- PO10. Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team and to manage projects and in multidisciplinary environments.
- PO11. Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8).
- PO12. Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.



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In addition to incorporating the above-listed POs, the educational institution may include additional outcomes in its learning programs. An engineering program that aims to attain the POs mentioned above must ensure that its curriculum encompasses all the attributes of the Knowledge Profile (WK1 – WK9) presented in Table 6.1 (Section 6.0) and as included in the PO statements. The ranges of Complex Problem Solving (WP1 – WP7) and Complex Engineering Activities (EA1 – EA5) are given in Tables 6.2 and 6.3 (Section 6.0), respectively.

The required attributes under this criterion are described in the following sub-criteria.

- i. The process involved in defining the POs must be described. POs specified by the program must be significantly equivalent to the twelve graduate attributes or POs of BAETE.
- ii. Each course must have clear quality requirements and facilitate the achievement of course outcomes (COs) through teaching-learning and assessment methods.
- iii. The program must demonstrate the availability of the lecture plan, including COs, course content, books, grading policy, assessment tools and samples of corresponding student works, and assessment of CO attainment.
- iv. Students' academic performance must be continuously monitored to ensure the achievement of outcomes.
- v. The program must demonstrate, using direct methods, that the students attain all POs by graduation.

5.3 Curriculum and Teaching-Learning Processes

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The curriculum must satisfy the relevant program-specific criteria.
- ii. The breadth and depth of the curriculum must be appropriate for solving complex engineering problems.
- iii. The curriculum must contain adequate courses on mathematics, natural sciences, computing, humanities, social sciences, and other non-engineering subjects.
- iv. Curriculum and teaching-learning processes must support the attainment of PEOs.
- v. The correlation between the course outcomes (COs) and POs must be appropriate.
- vi. How each attribute of the Knowledge Profile (WK1 – WK9) is addressed in the curriculum must be demonstrated through mapping. The program must also demonstrate how the attributes of the Range of Complex Engineering Problems (WP1 – WP7) and Complex Engineering Activities (EA1 – EA5) are incorporated into the teaching, learning, and assessment. Additionally, the program must demonstrate how various United Nations Sustainable Development Goals are considered in teaching, learning, and assessment.
- vii. The teaching-learning processes and assessment tools selected for each course must be effective and appropriate for achieving the relevant outcomes, including those pertaining to complex engineering problems and activities, if applicable.

5.4 Interactions with the Industry

A communication channel between the educational institution and the industry should exist. The industry should be encouraged to provide feedback concerning the quality of the teaching-learning process. Requirements under this criterion are described in terms of the following sub-criteria.



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- i. The industry must participate in developing the curriculum to ensure that it is relevant, regularly updated, and meets the needs of the industry, particularly in areas experiencing rapid changes.
- ii. The program must provide students with the opportunity to obtain industrial experience.

5.5 Continuous Quality Improvement

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The program must have a functional quality assurance system under an institutional framework.
- ii. The program must address the concerns, weaknesses, and deficiencies the BAETE evaluation team identified in the last review.
- iii. The students receive periodic feedback on their academic performance from the course teachers.
- iv. The findings of the CQI exercises for PEOs must be evaluated, and the identified shortcomings and limitations must be used regularly to refine and improve the program.
- v. POs must be assessed on a regular cycle. The program must undertake regular appropriate review of POs, considering feedback from relevant stakeholders.
- vi. The program must evaluate the curriculum and teaching quality regularly while considering feedback from faculty members, students and other stakeholders, and use the results of these evaluations for continuous improvements.

5.6 Students

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The program must have and practice published policies for the admission and transfer of students into the program.
- ii. The admission requirements must be appropriate for selecting students with the potential to achieve the program outcomes, including good grades in mathematics and natural sciences.
- iii. Every student must be assigned an advisor. The advisor must counsel, guide, and mentor the student on all academic and professional matters.
- iv. Students' workload must enable them to participate in extra- and co-curricular activities and the activities of relevant professional societies.

5.7 Faculty

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The department must have a sufficient number of full-time faculty members to ensure that they are not overloaded with courses and that the program does not become overly dependent on part-time faculty members.
- ii. The department must have a good blend of senior and junior faculty members with adequate academic qualifications and specializations in areas closely related to the program.
- iii. The teacher-student ratio, class size, and teaching load must not compromise opportunities for interactions between the students and faculty members.
- iv. Faculty members must be responsible for designing and updating the curriculum, establishing course and program outcomes, and selecting and using appropriate assessment tools for evaluating student performance in classes and the achievement of outcomes.
- v. Faculty members must be engaged in research, development and/or professional activities.



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- vi. All the faculty members must be adequately trained to establish course outcomes, conduct teaching-learning activities that are appropriate for the outcomes and assess the level of outcome achievement.

5.8 Governance, Finance and Safety

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The officers and statutory committees of the institution must function effectively as per their roles defined in the institutional act/statute.
- ii. The administrative and academic policies must be put into practice.
- iii. The financial resources committed to the program must be adequate for the appropriate functioning of the program, including recruiting and retaining qualified faculty members and procuring the necessary lab equipment and tools to support teaching and learning.
- iv. A safety plan must exist and be practiced to address the risk from manmade or natural hazards, including fire detection and suppression, as well as incidents and accidents in the laboratories.

5.9 Academic Facilities and Technical Support

Requirements under this criterion are described in terms of the following sub-criteria.

- i. The institution must have a well-stocked library with adequate books, e-books, journals, and other relevant resources for the program.
- ii. The learning environment must be adequate and conducive.
- iii. Every student must have the opportunity to work in the laboratories.
- iv. Students and faculty members must have access to adequate computing and Internet facilities, including hardware, software tools, and support.

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6.0 TABLES

Table 6.1: Knowledge and attitude Profile

	Attribute
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

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Table 6.2: Range of Complex Engineering Problem Solving

Attribute	Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7:
Depth of knowledge required	WP1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	WP2: Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	WP3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	WP4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	WP5: Address problems not encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	WP6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
Interdependence	WP7: Address high level problems including many components or sub-problems that may require a systems approach

Table 6.3: Range of Complex Engineering Activities

Attribute	Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:
Range of resources	EA1: Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
Level of interactions	EA2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
Innovation	EA3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
Consequences to society and the environment	EA4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5: Can extend beyond previous experiences by applying principles-based approaches



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7.0 THE BAETE ACCREDITATION MANUAL, STANDARD OPERATING PROCEDURES, TEMPLATES, FLOWCHARTS, SCHEDULE, AND WORK INSTRUCTIONS

See ACC-MAN-00 for details.

8.0 REVISION HISTORY

Date	Version	Description of Change
23 March 2017	1.0	Initial Release
05 March 2019	2.0	Year-long Final Year Design Project or Capstone Project is no longer obligatory in the curriculum. Demonstration of culminated POs in solving complex engineering problems remains required, and the year-long Final Year Design Project or Capstone Project is the preferred method. The descriptions of POs have been updated following IEA Version 3.21, June 2013. Requirements of the Knowledge Profiles (K1-K8) have been explicitly mentioned in the description of each PO. The tables describing the Range of Complex Engineering Problem-Solving and Range of Complex Engineering Activities have been updated following IEA Version 3.21, June 2013. The program should map out how each attribute of the Knowledge Profile (K1 – K8) is addressed in the curriculum. The program should also demonstrate how each attribute of the Range of Complex Engineering Problems (P1 – P7) and Complex Engineering Activities (A1 – A5) is incorporated in teaching, learning and assessment. Results of evaluation of PO attainment should be reported.
16 May 2022	2.1	The required and desired attributes of each of the ten criteria have been listed in terms of sub-criteria. The required attributes are indicated through the word must, while the desired attributes are indicated through the word should. Section 4.8: Titles for the POs have been removed. Reference to Annex III Evaluation Team Report (ETR) template has been added in the first paragraph of chapter 4.
23 April 2024	2.2	The purpose and scope of the document were added; roles and responsibilities were added; definitions and acronyms were added; there were no changes in the previous content.
23 April 2024	3.0	Released as a standalone document; desired attributes are removed from the criteria; criteria are rearranged, and criteria relating to outcomes are brought forward; Organization and Governance (criterion 1 in v2.1) and Financial and Physical Resources (criterion 2 in v2.1) were merged as “Governance, Finance and Safety”; sub-criteria under each criterion were reviewed and revised.